High-level Conceptual Design of RF Communication System at South China Sea

Sally Law

Embry-Riddle Aeronautical University

July 5, 2026

Table of Contents

1.0 Introduction to the use of RF communication system in South China Sea

As one of the world's most highly contested waterways, the South China Sea has become a continuous arena for intensive Electronic Warfare (EW), where regional state actors utilize "salami-slicing" and "cabbage tactics" to assert dominance entirely within the electromagnetic spectrum rather than through kinetic conflict. Consequently, the design of the UAV RF communication system must account for RF interference and tactical datalink interception, as discussed in the contingency plan. To successfully establish a robust communication link in this hostile theater, the baseline link budget calculation must withstand environmental degradation to guarantee a sufficient fade margin for survival. As the South China Sea functions as a contested grey zone where standard ITU databases do not always accurately reflect reality, the design of the UAV communication system must heavily rely on Open-Source Intelligence (OSINT) and geopolitical risk monitoring is mandatory prior to deployment. With the intended UAV L-band radio wave operating at 1060.5 MHz, a frequency highly sensitive to the tropical maritime boundary layer of the SCS, the link budget must dynamically account for the following constraints:

- **Rain & Moisture Absorption:** Heavy tropical downpours and high absolute humidity cause localized signal attenuation and hydrometeor scattering.
- **Ionospheric Turbulence:** Solar activity and ionospheric scintillation can induce rapid phase and amplitude fluctuations, degrading signal coherence.

***Note**: While additional environmental losses can be factored in, the professor has indicated that only these two specific environmental losses will be required for this class project.*

1.1 UAV Operational Frequency Validation

Operational frequency validation is critical to bridge the gap between theoretical regulatory compliance and the hostile electromagnetic realities of the South China Sea. This process ensures that while the UAV's 1060.5 MHz operating frequency aligns with international framework allocations, the system design actively accounts for de facto electronic warfare, contested spectrum sovereignty, and severe out-of-band interference risks near military air-defense channels.

1.2 FCC/ITU Frequency Zoning Validation at South China Sea

To initiate the operational frequency validation for the target 1060.5 MHz link, the target frequency of 1060.5 MHz was cross-referenced against the International Telecommunication Union (ITU) framework for region 3 to establish a baseline of regulatory legitimacy. The ITU frequency allocation matrix confirms that the band surrounding 1060.5 MHz is strictly reserved for air-to-ground communications, aligning precisely with its primary international designation for Aeronautical Radionavigation Service (ARNS). This preliminary regulatory alignment ensures a theoretically low-risk civilian interference profile, completely isolated from standard commercial networks.

Figure 1: *ITU Frequency Zoning region*

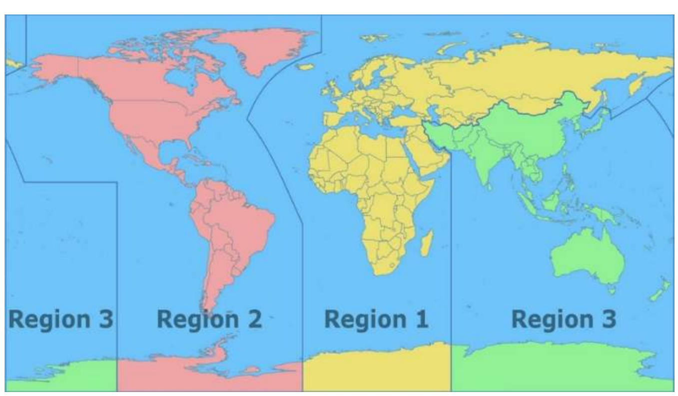


Table 1: *ITU Frequency Allocation Table for Frequency between 894-1400 MHz*

Table of Frequency Allocations			894-1400 MHz (UHF)		Page 31
International Table			United States Table		
Region 1 Table	Region 2 Table	Region 3 Table	Federal Table	Non-Federal Table	FCC Rule Part(s)
960-1164 Aeronautical Mobile (R) 5.327A Aeronautical Radionavigation 5.328 5.328AA			960-1164 Aeronautical Mobile (R) 5.327A Aeronautical Radionavigation 5.328 5.328AA US78 US224		Aviation (87)

Table 2: *Footnote from ITU specifying the use for frequency range at 1060.5 MHz*

(i) 5.327A The use of the frequency band 960-1164 MHz by the aeronautical mobile (R) service is limited to system that operate in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 417 (Rev.WRC-15)

(328) 5.328 The use of the band 960-1215 MHz by the aeronautical radionavigation service is reserved on a worldwide basis for the operation and development of airborne electronic aids to air navigation and any directly associated ground-based facilities.

1.3 Frequency Spectrum Sovereignty

While regulatory compliance mandates cross-referencing spectrum allocations with ITU Region 3 guidelines, regulatory validation alone is insufficient for real-world operations within the South China Sea. Due to contested sovereignty claims driven by the Nine-Dash Line, regional state actors explicitly reject international maritime boundaries and legal frequency zoning across roughly 90% of the theater. Consequently, physical spectrum usage is governed by forward-deployed, *de facto* military enforcement rather than *de jure* international law. Because the South China Sea functions as a contested grey zone where standard ITU databases do not accurately reflect reality, transitioning from passive regulatory compliance to active Open-Source Intelligence (OSINT) and geopolitical risk monitoring is mandatory prior to operation.

1.4 Tactical Conflict and Spectral Risk

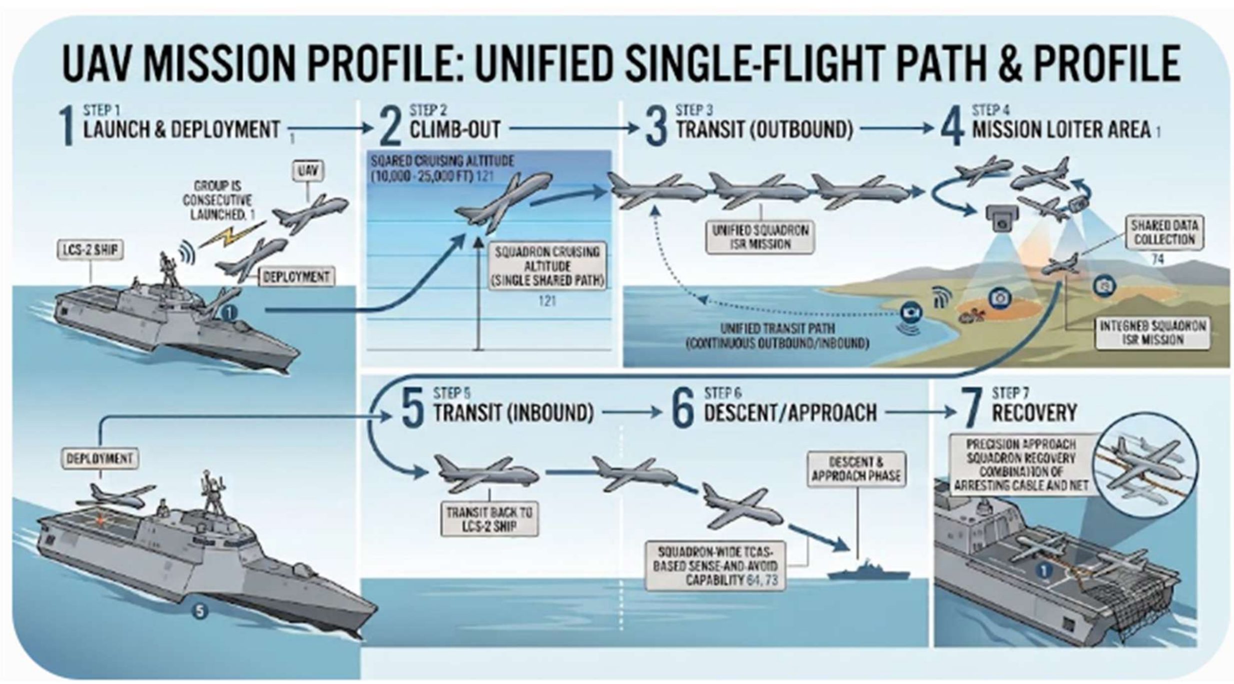
Operating a custom UAV platform at 1060.5 MHz introduces a tactical risk due to its extreme proximity to the core frequencies of military situational awareness at 1030 MHz (surface interrogators) and 1090 MHz (airborne transponders). In the highly volatile South China Sea theater, major military actors, including the PLAN, the US Navy, and regional coastal forces, depend entirely on these adjacent channels for mission-critical Identification Friend or Foe (IFF) and Secondary Surveillance Radar (SSR) systems. Deploying a high-power UAV transmitter in this narrow window presents an acute danger of out-of-band emissions bleeding directly into these strict guard bands. Such spectral encroachment could inadvertently degrade, blind, or disrupt active military air-defense and collision-avoidance tracking, risking catastrophic misidentification or an unintended kinetic escalation in a heavily contested grey zone. Thus, the UAV communication system's operating band must be rigorously filtered to enforce strict spectral containment, preventing out-of-band emissions from bleeding into adjacent channels and causing destructive inter-channel interference (ICI).

2.0 Mission Profile for Communication System & Navigation Plan

The SMEM flight profile utilizes Line-of-Sight (LOS) communications as the primary link during launch, climb-out, descent, and recovery, while also serving as an active redundant backup whenever the aircraft is within operational physical range of a suitably equipped ship. For long-range operations, Beyond Line-of-Sight (BLOS) communication serves as the primary command-and-control link throughout the transit ingress, mission loiter, and transit egress phases. Under this dual-link framework, the system switches to BLOS for mission execution while reverting to automated or pilot-controlled LOS links during critical launch and shipboard recovery operations.

Since the launch platform is an LCS-2 ship has a uniquely wide trimaran flight deck rather than a conventional longitudinal runway, a VTOL or helicopter configuration was selected and reflected in the Navigation Plan (NavPlan).

Figure 2: *UAV Operation Mission Profile*



3.0 Introduction to UAV Line-of-Sight (LOS)

To ensure the UAV can maintain continuous line-of-sight during operation, both the geometric Line-of-Sight (LOS) path and the L-Band radio frequency (RF LOS) link were validated using MATLAB terrain datasets to confirm zero environmental or signal obstructions.

The following analysis successfully validated requirements [UAV-2] and [UAV-3], serving as the formal technical verification artifacts required to sell off the RF communication system design.

3.1 UAV Navigation Path Design & Fresnel Zone clearance validation

The 10 NM waypoint profile was designed using MATLAB and Earthstar Geographics terrain datasets, confirming zero physical line-of-sight (LOS) obstructions between the Littoral Combat Ship and the UAV during critical near-ship phases. Furthermore, geometric analysis verifies complete first Fresnel Zone clearance. To validate Fresnel Zone clearance between the two offshore points, calculate the first Fresnel Zone radius at the midpoint of the link based on your operating frequency, and ensure the line-of-sight path clears the ocean surface by at least 60% of that radius plus an allowance for earth curvature and atmospheric refraction. With the UAV cruising at 24,000 ft, the maximum 28.3-meter Fresnel radius remains thousands of feet above the sea surface, eliminating any risk of multipath encroachment or diffraction losses.

Figure 3: *UAV flight path route map (Zoom In)*

Figure 4: *UAV flight path route map (Zoom out)*

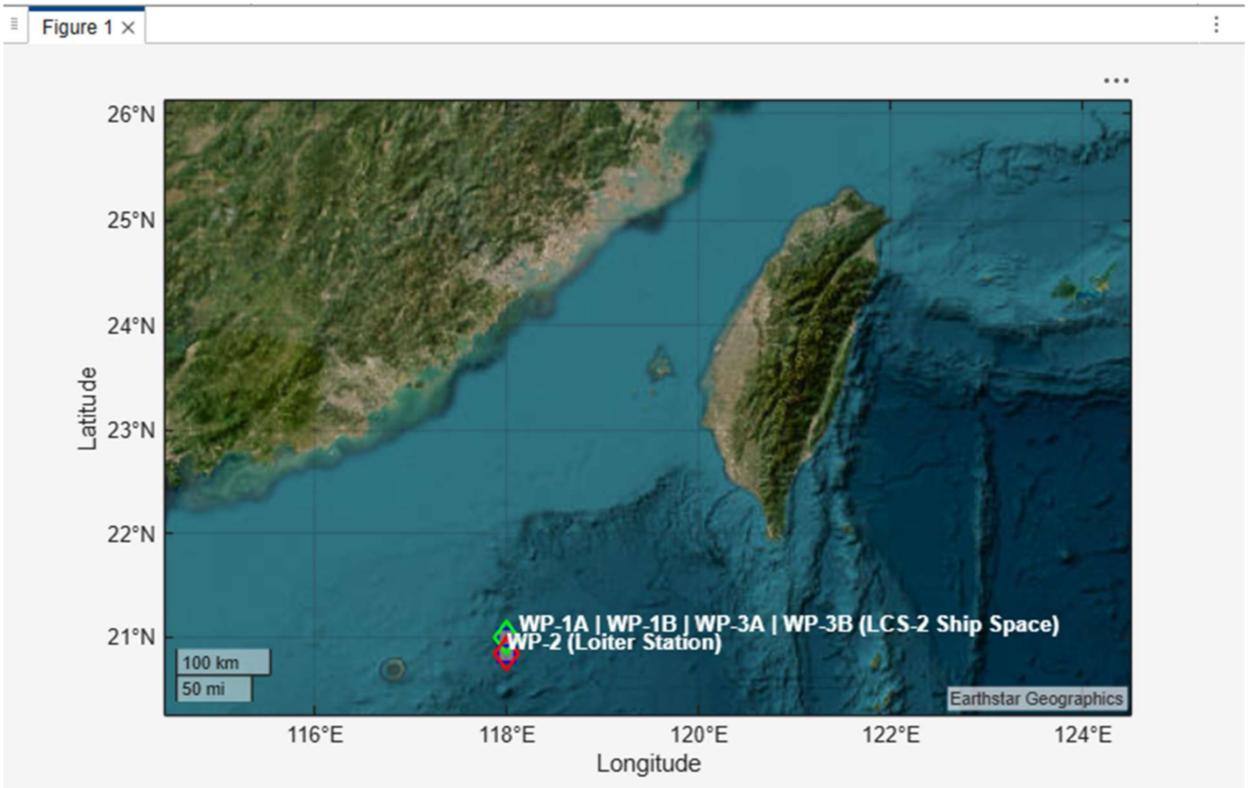


Table 3: *UAV VTOL Navigation Flight Plan (NavPlan)*

WayPT	Type / Phase	Latitude	Longitude	Altitude (MSL)	Distance from Ship	Description
WP-1A	Launch / Origin	21°00'00" N	118°00'00" E	Sea Level (0 ft)	0 NM	deployment from LCS-2 ship via direct RF LOS command and control (C2); BLOS is also available
WP-1B	Top of Vertical Climb (TOVC)	21°00'00" N	118°00'00" E	24,000 ft	0 NM (Over Ship)	Climb-out to a cruising altitude at 24000 ft; maintain LOS C2 control; BLOS served as redundant
WP-2	Loiter Station (AOI)	20°50'00" N	118°00'00" E	24,000 ft	10.0 NM South	Transit to ISR mission site (Entering Area of Interest AOI), station directly above Mission Loiter (ISR); mission collect; maintain LOS
WP-3A	Inbound Return / End of Forward Transit (EOFT)	21°00'00" N	118°00'00" E	24,000 ft	0 NM (Over Ship)	Transit to LCS-2 ship; maintain RF LOS
WP-3B	Vertical Descent / End of Flight (EOF)	21°00'00" N	118°00'00" E	Sea Level (0 ft)	0 NM	Descent/approach; Maintain direct RF LOS C2 Control; BLOS remains available

3.2 RF LOS Communication Design (Primary Communication)

To establish and continuously maintain communication link, an L-Band RF Line-of-Sight link requires an unblocked, straight physical path between the ship's directional dish antenna and the UAV. As the updated flight profile shows, the loiter station (WP-2) sits at 20°50'00" N, 118°00'00" E, which is a downrange distance of exactly 10 Nautical Miles from the launch vessel. Because the designated operational height of the UAV is 24,000 ft, the physical curvature of the Earth does not drop the ship below the horizon relative to the aircraft; therefore, the entire flight profile physically stays well within the geometric line-of-sight footprint of the LCS-2 ship. Mathematically speaking, the horizon at this altitude extends past 130 Nautical Miles, meaning it is not a risk of an absolute loss of direct radio line-of-sight due to planetary curvature during this short-range mission. To strictly comply with the mandated mission profile requirements, a programmatic handover to the satellite-based Beyond Line-of-Sight (BLOS) architecture may be executed during the transit and loiter phases.

Additionally, a pre-flight RF spectrum scan will be performed to analyze active frequency utilization within the operational area, allowing for the application of dynamic frequency notching if necessary to mitigate localized signal interference and preserve link integrity.

The UAV RF operational profile validates design requirements against [UAV-8], [UAV-15] and [UAV-20].

3.3 UAV Beyond Line-of-Sight (BLOS) Design (Redundant Communication)

RF Beyond Line-of-Sight (BLOS) communication serves as the redundant command-and-control link throughout the transit ingress, mission loiter, and transit egress phases. Under this dual-link framework, the system switches to BLOS for mission execution while reverting to automated or pilot-controlled LOS links during critical launch and shipboard recovery operations. ***Note:*** *As stated in the customer specification (project prompt), analyzing the Beyond Line-of-Sight (BLOS) link budget is outside the scope of this project.*

4.0 Link Budget Design (RF LOS Communication)

4.1 Signal Link Margin

In radio frequency (RF) communication systems, the signal link margin represents the safety buffer between the actual received signal strength and the minimum power required by the receiver to maintain a reliable connection. This baseline ensures that the communication link remains robust against unexpected environmental losses, atmospheric attenuation, or changes due to environmental changes. The net signal margin is defined as:

$$\boxed{\text{Net signal margin (dB)} = \text{Available SNR (dB)} - \text{Required SNR (dB)}}$$

In Unmanned Aircraft System (UAS) communication link, the Required SNR is the minimum threshold of signal clarity that the receiver must capture to successfully demodulate and decode incoming data. If the actual signal quality remains above the Required SNR, data packets are processed reliably. Conversely, if the signal quality drops below this threshold, the system experiences packet loss, command dropouts, and video telemetry degradation. The

Required SNR accounts for both digital modulation requirements and hardware bandwidth constraints, and is calculated as follows:

$$\boxed{Required\ SNR\ (dB) = \frac{E_b}{N_0} + Data\ Rate\ (dBHz) - Bandwidth(dBHz)\ \text{(eq. 1)}}$$

Converting the operational data rate and channel bandwidth into logarithmic terms yields:

$$Data\ Rate\ (dBHz) = 10log_{10}(R_{data}) = 10log_{10}(138kbps) = 51.40\ dBHz\ (2)$$

$$Bandwidth(dBHz) = 10log_{10}(B) = 10log_{10}(120kbps) = 50.79d\ BHz\ (3)$$

By substituting (2) and (3) into (eq. 1),

$$Required\ SNR = \frac{E_b}{N_0} + Data\ Rate\ (dBHz) - Bandwidth(dBHz)$$

$$Required\ SNR = 3.5\ dB + 51.40\ dBHz - 50.79\ dBHz$$

$$\boxed{\boldsymbol{Required\ SNR = 4.11\ dB}}$$

Per the system requirements defined in [UAV-1], the UAS communication link must maintain a target Net Signal Margin of at least 31.5 dB to ensure stable link connectivity throughout the mission profile. To determine the minimum Available SNR needed to satisfy this requirement, the margin equation is rearranged as follows:

Net signal margin (dB) = Available SNR (dB) – Required SNR (dB)

31.5 dB = Available SNR (dB) – 4.11 dB

Available SNR (dB) = a minimum of 35.61 dB

Thus, to guarantee robust command-and-control capabilities, the UAV communications subsystem shall maintain an Available Signal-to-Noise Ratio (SNR) of at least 35.61 dB at the receiver, ensuring the system satisfies the mandatory 31.5 dB link margin. For the analytical assessments detailed in subsequent sections, the Available SNR is defined as:

Available SNR(dB) = Effective carrier power (C_{eff})(dBm) - Effective noise power (N_0)(dBm)

Here's the summary of the minimum SNR for UAV to establish link with a minimum of 31.5 dBm link margin:

Table 4: *UAV Communication System SNR Summary*

		Operation	Units	Value
Summary	Available SNR		dB	35.61
	Required SNR		dB	4.11
	Net signal margin (link margin)		dB	31.50

4.2 Effective Noise Power (N_0)

Effective Noise Power represents the total raw power of the background electrical static present within the receiver's active channel, combined with the internal noise introduced by the receiver's electronics. This value establishes the absolute noise floor of the communication system; the incoming target signal must arrive with sufficient power above this floor to be successfully detected and demodulated. The effective noise power scales the baseline thermal noise density relative to the system's operational bandwidth and hardware quality, calculated as:

Effective Noise Power (dBm) = Thermal Noise Density (kT) (dBW/Hz) + Rx Bandwidth ($10\log_{10}B$) + Noise Figure (NF) (dB)

Driven by thermodynamic principles, the thermal noise density (N_0) is the baseline electrical static caused by the thermal agitation of electrons. Assuming a standard operational ambient temperature of 290 K, the baseline thermal noise density is expressed as:

Thermal Noise Density (N_0) (dBW/Hz) = $10\log_{10}(k) + 10\log_{10}(T)$ = -174 dBm/Hz

$$\boxed{\text{Thermal Noise Density } (N_0) \text{ (dBW/Hz)} = -174 \text{ dBm/Hz}}$$

Given Ideal Receiver Bandwidth, B=120 kHz,

$$\boxed{\text{Rx noise bandwidth (BW) (dBHz)} = 10\log10(B) = 50.79 \text{ dBHz}}$$

The receiver hardware introduces an additional degradation factor, specified by a Noise Figure (NF) of 4dB.

$$\boxed{\textbf{Noise Figure = 4 dB}}$$

Consequently, the absolute receiver noise floor for this configuration is:

Effective Noise Power = Thermal Noise Density (kT)+Rx Bandwidth ($10\log_{10}B$)+Noise Figure (NF)

Effective Noise Power = -174 dBm/Hz – 50.79 dBHz – 4 dB

$$\boxed{\textbf{Effective Noise Power } (N_0) = \textbf{-119.21 dBm}}$$

Here's the summary effective noise power for this UAV mission:

Table 5: *UAV Communication System Effective Noise Power Summary*

		Operation	Units	Value
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00
	Rx noise bandwidth, BW	-	dBHz	50.79
	Rx noise figure, NF	-	db	4.00
	Effective noise power	-	dBm	-119.21

4.3 Link Budget Losses

To maintain reliable command-and-control operations in the challenging South China Sea environment, the link budget incorporates critical signal degradation factors and environmental margins to ensure system functionality under extreme conditions.

4.3.1 Free Space Path Loss (FSPL)

Free Space Path Loss (FSPL) is the reduction in power density of an electromagnetic signal as it travels in a straight line through an unobstructed, ideal vacuum, completely free of obstacles, absorption, or reflections. To calculate FSPL, the standard equation is:

$$FSPL(dB) = 20log_{10}(d) + 20log_{10}(f) + 20log_{10}(\frac{4\pi}{c})$$

By assuming worst case scenario, where:

Frequency (f) = 1060.5 MHz

Horizontal Ground distance = 10 NM

UAV altitude (h) (worst case) = 25000 ft = 25000 ft / (6076.12 ft/NM) = 4.115 NM

True Slant Range (d) can be calculated by:

$d = \sqrt{(10NM)^2 + (4.115NM)^2}$ =10.814 NM (1.852km/NM) = 20.028 km

Free Space Path Loss:

$$FSPL(dB) = 20log_{10}(20.028km) + 20log_{10}(1060.5MHz) + 32.45$$

FSPL (dB) = 118.31 dB

4.3.2 Atmospheric Absorption Loss

Atmospheric loss, or atmospheric attenuation, represents the reduction in electromagnetic signal power as it propagates through the Earth's atmosphere. This loss can be modeled using an empirical power-law formula. To account for these environmental variations across the SMEM mission profile, the following fixed atmospheric loss is integrated into the downward line-of-sight link budget:

Atmospheric Loss, 0.926 dB (est.)

4.3.3 Precipitation Losses

Precipitation loss, or rain attenuation, describes the degradation of electromagnetic signal strength due to atmospheric moisture such as rain, snow, hail, or fog. These losses are calculated using the empirical framework defined in ITU-R P.838-3, *Specific attenuation model for rain for use in prediction methods.* To evaluate the environmental margins for this mission profile, the following estimated precipitation loss is factored into the line-of-sight link budget:

Precipitation loss, 0.02 dB (est.)

4.3.4 Total RF Propagation Loss

Thus, the total propagation losses is:

$$\boxed{\boldsymbol{L_{Loss} = FSPL(dB) + L_{Atmospheric} + L_{Rain} = -119.26\ dB}}$$

Here's the summary of the RF Link Loss for this UAV mission:

Table 6: *UAV Communication System RF Link Loss Summary*

		Operation	Units	Value
Propagation	Free space path loss	+	dB	-118.31
	Atospheric absorption, L_P,Atm	+	dB	-0.93
	Precipitation absorption, L_P, Precip	+	dB	-0.02
	Total Propagaton Loss		dB	-119.26

4.4 Effective Carrier Power

Effective Carrier Power (C_{eff}) is the total magnitude of signal power that successfully arrives at the receiver's demodulation after accounting for all cumulative gains and systemic losses along the transmission path. For a downlink transmission, the Effective Carrier Power arriving at the Ground Control Station (GCS) receiver is mathematically modeled as:

Effective Carrier Power(C_{eff})(dBm) = EIRP(dBm) + Total Propagation Loss(dB) + Total G_{RX} (dBi) – Total L_{RX}(dB)**(eq. A)**

From section 3.5.1, the link requires a minimum Available SNR of 35.61 dB to preserve the mandatory safety margin, and the relationship between signal quality, carrier power, and the receiver noise floor is defined as:

Available SNR (dB) = Effective carrier power (dB) - Effective noise power (dBm),

Therefore,

Effective carrier power (min.)(dBm) = Available SNR (min.)(dB) - Effective noise power (dBm)

Minimum Effective carrier power (dBm) = 35.61 dB + (-119.21 dBm)

Minimum Effective carrier power (C_{eff}) = - 83.60 dBm (i)

Here's the summary of the minimum Effective Carrier Power requirement for UAV to establish link with a minimum of 31.5 dBm link margin:

Table 7: *UAV Communication System Effective Carrier Power Summary*

		Operation	Units	Value
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00
	Rx pointing Loss, L_R,Point	+	dB	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00
	Effective carrier power		dBm	-83.60

4.5 Equivalent Isotropic Radiated Power (EIRP)

Equivalent Isotropically Radiated Power (EIRP) represents the total effective power radiated by an antenna in its direction of maximum gain, measured relative to a theoretical isotropic antenna that radiates power equally in all directions. EIRP accounts for transmitter power output, internal transmission line losses, and the intentional directive gain of the antenna.

From Section 5.3, the minimum required Effective Carrier Power (C_{eff}) arriving at the receiver is -83.60 dBm. To isolate the minimum required EIRP from the primary link budget equation, the system parameters are arranged as follows:

$$\boxed{EIRP(dBm) = P_{TX} - L_{cable} + G_{antenna}}$$

From section 3.5.4, ground station receiver characteristics and path parameters are defined as:

RX polarization loss = -3 dB, Line Loss = -1 dB, and total propagation loss is -119.26 dB (from section 3.5.2)

By substituting (i) into (eq. A) from section 3.5.4,

$$-83.60 \text{ dBm} = \text{minimum EIRP} + \text{Total Propagation Loss} + \text{Total } G_{RX} - \text{Total } L_{RX}$$

$$-83.60 \text{ dBm} = \text{minimum EIRP} + (-119.26 \text{ dB}) + (19.0 \text{ dBi}) + (-3.0 \text{ dB} - 1.0 \text{ dB})$$

$$\boxed{\textbf{minimum EIRP} = \textbf{20.66 dBm}}$$

Thus, the airborne transmitter configuration must achieve a minimum EIRP of 20.66 dBm to close the link under worst-case operational conditions.

Additionally, the link budget can also be analyzed from the perspective of the airborne transmitter components using the standard EIRP definition:

$$\boxed{EIRP(dBm) = P_{TX} - L_{cable} + G_{antenna},}$$

To reconcile the actual hardware performance with the required baseline of 20.66 dBm, the minimum required transmitter antenna gain (G_{TX}) must satisfy:

$$minimum\ P_{TX} \geq EIRP(dBm) + L_{cable} - G_{antenna}$$

$$\boxed{\boldsymbol{minimum\ G_{Antenna} \geq 7.66\ dBi}}$$

$$\text{Linear Antenna Gain} = 10^{\frac{7.66\ dBi}{10}} = 5.834$$

The transmitted power output at the output port of the UAV RF transmitter is 7.66 dBi, or 5.834.

Here's the summary of the minimum EIRP requirement for UAV to establish link with a minimum of 31.5 dBm link margin:

Table 8: *UAV Communication System EIRP Summary*

		Operation	Units	Value
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00
	Tx component line losses, L_T,Line	+	dB	-1.00
	Tx antenna gain, Gt **(minimum)**	+	dBi	7.66
	Tx pointing loss, L_T,point	+	dB	0.00
	Tx radome loss, L_T,radome	+	dB	-1.00
	EIRP (minimum)		dBm	20.66

4.6 Required Antenna Gain Adjustment Table

Since the UAV communication system must maintain a strict minimum realized gain of 7.66 dBi to close the link budget with your required 31.5 dB margin, you must over-design the antenna's physical structure, the following UAV antenna must follow the specification below based on its efficiency rating:

Table 9: *Adjusted Antenna Gain due to Antenna Efficiency Factor operating at 290K:*

Antenna Efficiency (%)	Required Minimum Realized Gain (dBi)	Internal Heat Loss (dB)	**Required Designed Base Gain (dBi)**	Equivalent Linear Gain Ratio
100%	7.66 dBi	0.00 dB	**7.66 dBi**	5.834
95%	7.66 dBi	0.22 dB	**7.88 dBi**	6.138
90%	7.66 dBi	0.46 dB	**8.12 dBi**	6.486
85%	7.66 dBi	0.71 dB	**8.37 dBi**	6.871
80%	7.66 dBi	0.97 dB	**8.63 dBi**	7.294

4.7 Minimum requirements for UAV Communication System (Link Budget)

Here is the summary of the minimum requirement to establish UAV communication link with a minimum net link margin of 31.50 dB:

Table 10: *Link Budget meets minimum net link margin of 31.50 dB*

Parameters			
	Frequency	1060.5	MHz
	Modulation Data Rate (Rdata)	138000	bits/s
	Bandwidth, B	120000	Hz
	Wavelength	0.927487	ft
	Tx Power	15	dBm
	UAS Antenna Gain **(minimum)**	7.66	dBi
	Control Station Antenna Gain	19	dBi
	Range, R	10	NM
	Required Eb/No	3.5	dB

		Operation	Units	Value
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00
	Tx component line losses, L_T,Line	+	dB	-1.00
	Tx antenna gain, Gt **(minimum)**	+	dBi	7.66
	Tx pointing loss, L_T,point	+	dB	0.00
	Tx radome loss, L_T,radome	+	dB	-1.00
	EIRP (minimum)		dBm	20.66
Propagation	Free space path loss	+	dB	-118.31
	Atospheric absorption, L_P,Atm	+	dB	-0.93
	Precipitation absorption, L_P, Precip	+	dB	-0.02
	Total Propagaton Loss		dB	-119.26
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00
	Rx pointing Loss, L_R,Point	+	dB	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00
	Effective carrier power (minimum)		dBm	-83.60
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00
	Rx noise bandwidth, BW	-	dBHz	50.79
	Rx noise figure, NF	-	db	4.00
	Effective noise power	-	dBm	-119.21
Summary	Available SNR **(minimum)**		dB	35.61
	Required SNR		dB	4.11
	Net signal margin (link margin) (minimum)		dB	31.50

4.8 Link Budget Requirements Allocation (RF LOS)

To successfully establish and maintain continuous communication, an additional margin is added to the link budget to account for unforeseen RF losses. Consequently, the overall link margin requirements allocation is updated to 35 dB.

Table 11: *Link Budget Allocation*

Parameters			
	Frequency	1060.5	MHz
	Modulation Data Rate (Rdata)	138000	bits/s
	Bandwidth, B	120000	Hz
	Wavelength	0.927487	ft
	Tx Power	15	dBm
	UAS Antenna Gain	**11.155**	dBi
	Control Station Antenna Gain	19	dBi
	Range, R	10	NM
	Required Eb/No	3.5	dB

		Operation	Units	Value
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00
	Tx component line losses, L_T,Line	+	dB	-1.00
	Tx antenna gain, Gt	+	dBi	11.16
	Tx pointing loss, L_T,point	+	dB	0.00
	Tx radome loss, L_T,radome	+	dB	-1.00
	EIRP		dBm	24.16
Propagation	Free space path loss	+	dB	-118.31
	Atospheric absorption, L_P,Atm	+	dB	-0.93
	Precipitation absorption, L_P, Precip	+	dB	-0.02
	Total Propagaton Loss		dB	-119.26
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00
	Rx pointing Loss, L_R,Point	+	dB	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00
	Effective carrier power		dBm	-80.10
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00
	Rx noise bandwidth, BW	-	dBHz	50.79
	Rx noise figure, NF	-	db	4.00
	Effective noise power	-	dBm	-119.21
Summary	Available SNR		dB	39.11
	Required SNR		dB	4.11
	Net signal margin (link margin)		dB	**35.00**

And the Adjusted Antenna Gain due to Antenna Efficiency Factor operating at 290K is as follow:

Table 12: *Antenna Base Gain and Realized Gain*

Antenna Efficiency (%)	Required Minimum Realized Gain (dBi)	Internal Heat Loss (dB)	**Required Designed Base Gain (dBi)**	Equivalent Linear Gain Ratio
100%	11.15 dBi	0.00 dB	**11.155**	13.047
95%	11.15 dBi	0.22 dB	**11.378**	13.733
90%	11.15 dBi	0.46 dB	**11.613**	14.496
85%	11.15 dBi	0.71 dB	**11.861**	15.349
80%	11.15 dBi	0.97 dB	**12.124**	16.308

Based on the link budget analysis and the UAV antenna configuration with a minimum Peak Directive Gain(G_T) of 12.124 dBi and an efficiency of 80% is selected for this operation, yielding a net transmitter gain of 11.15 dBi, as specified in [CM-4].

4.9 Link Budget Prediction based on Hardware selection (RF LOS)

Following the selection of the UAV antenna, the RF Line-of-Sight (LOS) link budget was updated to predict baseline system performance. Please refer to Section 7: Hardware Selection Summary) for the hardware parameters used in this Link Budget Prediction update.

4.10 Link Budget Prediction results

The resulting predictive analysis, detailed below, verifies signal viability using the finalized hardware parameters, as defined and captured in *Section 7.1: Final System parameters*.

Table 13: *Link Budget Prediction results*

Parameters			
	Frequency	1060.5	MHz
	Modulation Data Rate (Rdata)	138000	bits/s
	Bandwidth, B	120000	Hz
	Wavelength	0.927487	ft
	Tx Power	15	dBm
	UAV Antenna Gain (Realized) GTX	**13.155**	**dBi**
	UAV Antenna Designed Base Gain (GTx)	**14.124**	**dBi**
	Anteanna Efficiency	**80**	**%**
	Control Station Antenna Gain	19	dBi
	Range, R	10	NM
	Required Eb/No	3.5	dB

		Operation	Units	Value
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00
	Tx component line losses, L_T,Line	+	dB	-1.00
	Tx antenna gain, Gt	+	dBi	13.16
	Tx pointing loss, L_T,point	+	dB	-0.10
	Tx radome loss, L_T,radome	+	dB	-0.50
	EIRP		dBm	26.56
Propagation	Free space path loss	+	dB	-118.31
	Atospheric absorption, L_P,Atm	+	dB	-0.93
	Precipitation absorption, L_P, Precip	+	dB	-0.02
	Total Propagaton Loss		dB	-119.26
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00
	Rx pointing Loss, L_R,Point	+	dB	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00
	Effective carrier power		dBm	-77.70
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00
	Rx noise bandwidth, BW	-	dBHz	50.79
	Rx noise figure, NF	-	db	4.00
	Effective noise power	-	dBm	-119.21
Summary	Available SNR		dB	41.51
	Required SNR		dB	4.11
	Net signal margin (link margin)		dB	**37.40**

5.0 Post-Link Budget Simulation, Analysis and Specification Updates

Following the finalization of the link budget design, allocation parameters, and performance predictions, the system analysis has been revised to reflect these completed baseline metrics. The comprehensive updates and corresponding technical evaluations are detailed in section 7: Final requirement specifications.

5.1 Post Update: UAV RF Coverage Analysis & RF LOS Re-validation

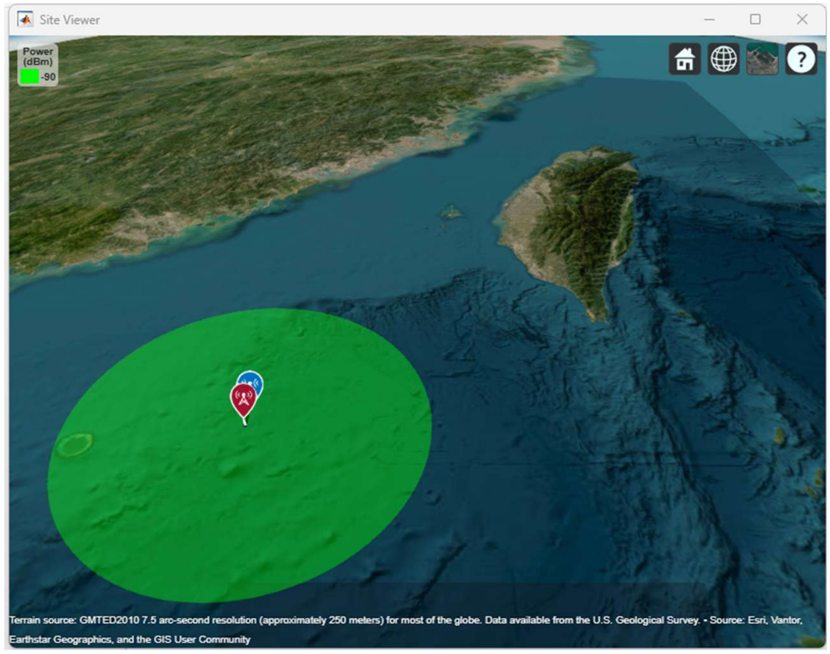
After finalizing the UAV communication system design, the updated antenna hardware specifications, waypoints from the navigation plan (NavPlan), and core design criteria are imported into MATLAB to execute the RF coverage analysis. Rain antennation and environmental losses are factored into the RF coverage analysis to model realistic environment. The simulation is generated using the following parameters:

```
% UAV TX Site
fq = 1.0605e9;                          % 1060.5 MHz
tx = txsite("Name","MissionLoiter", ...
    "Latitude",20.833333, ...
    "Longitude",118.00000, ...
    "Antenna",design(dipole,fq), ...    %"Antenna","isotropic", ...
    "AntennaHeight",7315.2, ...         % Units: meters (24,000 ft)
    "TransmitterFrequency",fq, ...      % Units: Hz
    "TransmitterPower",15);             % Units: Watts

% LCS-2 RX sites
rxNames = ["LCS-2"];
rxLocations = [21.0000 118.0000];       % LCS-2 Ground Control Station
rxSensitivity = -90;                % Units: dBm; S = kTB + NF + SNR(min) -115.10;

rxs = rxsite("Name",rxNames, ...
    "Latitude",rxLocations(:,1), ...
    "Longitude",rxLocations(:,2), ...
    "Antenna",design(dipole,tx.TransmitterFrequency), ...
    "ReceiverSensitivity",rxSensitivity); % Units: dBm
```

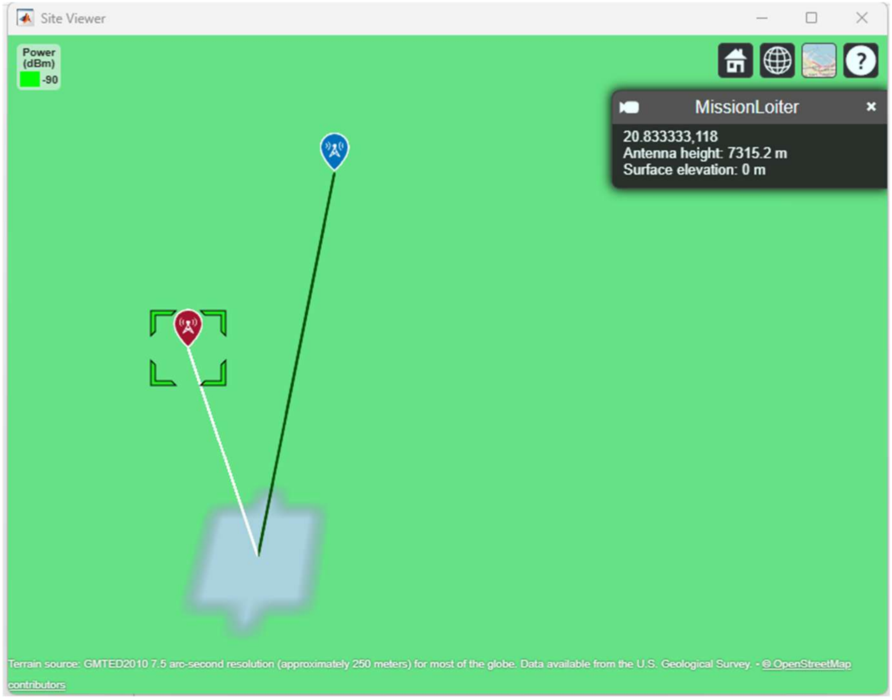
Figure 5: *UAV RF Coverage Analysis*



The MATLAB coverage analysis confirms that the system successfully establishes a robust RF link throughout the specified operational footprint, validating the viability of the communication design.

Figure 6: *UAV RF Coverage Analysis: LCS-2 View*

Figure 7: *UAV RF Coverage Analysis: Mission Loiter View*



6.0 Requirements Refinement & Final Requirements Specification

Following the finalization of the link budget design, allocation parameters, and performance predictions, the system requirements have been revised to reflect these design updates. The final requirements specification is detailed in the table below.

Table 14

UAV System Requirement Specification (SRS): Communication System Subset

Object ID	Object Type	Requirement	Traceability	Compliance Status
UAV System Requirement Specification (SRS)				
UAV Communication System				
UAV-1	Performance	The communication system shall maintain a minimum end-to-end link margin of 35.0 dB under worst-case operational and environmental conditions.	CM-1, CM-3, CM-4, CM-5, CM-6, CM-7, CM-8	Comply
UAV-2	Performance	The communication system shall achieve a link availability of no less than 97% over the designated geographic operational region.		Comply (Sec 5.1)
UAV-3	Performance	The communication system shall deliver user data at a maximum baseband Bit Error Rate (BER) of 1x10^-6 at the maximum operational range.	CM-2	Comply

Table 15

Communication subsystem specification

Object ID	Object Header	Requirement Text	Object Type	Ver. Method	Compliance Value	Ver. Evidence
UAV Communication Subsystem Specification						
RF subsystem Performance Requirements						
CM-1	Link Margin	The UAS communication subsystem shall maintain a minimum Net Signal Margin of 35.0 dB under worst-case operational conditions, including maximum operational range, adverse atmospheric attenuation, and worst-case aircraft attitude.	Performance	Analysis, Test	37.40 dB	Section 3: Link Budget Design
CM-2	Bit Error Rate	The ground station receiver demodulator shall successfully demodulate and decode incoming bitstreams with a maximum Bit Error Rate (BER) of 10-6 at a minimum received Eb/N0 threshold of 3.5 dB, inclusive of all internal modem implementation losses.	Performance	Analysis, Test	Eb/N0 3.5dB, BER 10-6 (Verified by design)	Section 3: Link Budget Design; Hardware specification
CM-3	Receiver Noise Bandwidth	The communication subsystem receiver shall operate with a total noise bandwidth (BW) of 50.79 dBHz +/- 0.15 dB (equivalent to 120 kHz).	Performance	Analysis, Test	50.79 dBHz (Verified by design)	Section 3: Link Budget Design; Hardware specification
UAV Transmitter Hardware Requirements						
CM-4	Transmitter Output Power	The UAV RF transmitter shall provide a minimum continuous conducted output power (P_TX) of 15.0 dBm at the output port.	Performance	Analysis, Test	15.0 dBm (Verified by design)	Section 3: Link Budget Design; Hardware specification

Object ID	Object Header	Requirement Text	Object Type	Ver. Method	Compliance Value	Ver. Evidence
CM-5	Transmitter Antenna Gain	The UAV transmitter antenna shall provide a minimum Peak Directive Gain (G_T) of 11.16 dBi with 80% antenna efficiency within the operational frequency band.	Performance	Analysis, Test	13.16 dBi	Section 3: Link Budget Design; Hardware specification
CM-6	Equivalent Isotropic Radiated Power (EIRP)	The integrated airborne subsystem shall achieve a minimum EIRP of 24.16 dBm.	Performance	Analysis, Test	25.56 dBm	Section 3: Link Budget Design
LCS-2 Ground Control Station Subsystem Requirements						
CM-7	Receiver Antenna Gain	The LCS-2 Control Station receiver antenna shall provide a peak directive gain (G_RX) of at least 19.00 dBi along the tracking boresight.	Performance	Analysis, Test	19.00 dBi (Verified by design)	Section 3: Link Budget Design; Hardware specification
CM-8	Receiver Noise Figure	The ground station receiver system Noise Figure shall not exceed 4.00 dB.	Performance	Analysis, Test	4.00 dB (Verified by design)	Section 3: Link Budget Design; Hardware specification
RF subsystem Functional Requirements						
CM-9	RF Interference	The communication subsystem shall account for multipath and maritime RF interference.	Functional	Inspection	-	Section 3: Link Budget Design
CM-10	Communication Standards	The communication subsystem shall be compatible with DO 362 L Band datalink standards.	Functional	Inspection	-	Section 3: Link Budget Design

7.0 Final Design Specification

The UAV communication system design has been successfully validated against the mission requirements for a VTOL-based ISR platform operating from an LCS-2 ship in the South China Sea. The link budget analysis, combined with hardware selection and terrain validation, confirms that the system is capable of maintaining a robust, reliable command-and-control link with a minimum net signal margin of 35.0 dB. The following section summarizes the final hardware specifications, link budget parameters, and key design decisions.

7.1 Final System Parameters

Table 16
Final System Parameters

Parameter	**Value**	**Units**
Operating Frequency (f)	1060.5	MHz
Modulation Data Rate (Rdata)	138,000	bits/s
Channel Bandwidth (B)	120,000	Hz
Wavelength (λ)	0.9275	ft
Transmitter Output Power (PTX)	15	dBm
UAV Antenna Gain (Realized) GTX	13.155	dBi
UAV Antenna Designed Base Gain (GTx)	14.124	dBi
UAV Antenna Efficiency (η)	80	%
LCS-2 Receiver Antenna Gain (GRX)	19	dBi
Operational Range (R)	10	NM
Required Eb/No	3.5	dB
System Noise Figure (NF)	4	dB

7.2 Final Link Budget Summary

Table 17

Final Link Budget (Design / Prediction) Summary

		Operation	Units	Value
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00
	Tx component line losses, L_T,Line	+	dB	-1.00
	Tx antenna gain, Gt	+	dBi	13.16
	Tx pointing loss, L_T,point	+	dB	-0.10
	Tx radome loss, L_T,radome	+	dB	-0.50
	EIRP		dBm	26.56
Propagation	Free space path loss	+	dB	-118.31
	Atospheric absorption, L_P,Atm	+	dB	-0.93
	Precipitation absorption, L_P, Precip	+	dB	-0.02
	Total Propagaton Loss		dB	-119.26
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00
	Rx pointing Loss, L_R,Point	+	dB	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00
	Effective carrier power		dBm	-77.70
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00
	Rx noise bandwidth, BW	-	dBHz	50.79
	Rx noise figure, NF	-	db	4.00
	Effective noise power	-	dBm	-119.21
Summary	Available SNR		dB	41.51
	Required SNR		dB	4.11
	Net signal margin (link margin)		dB	**37.40**

Table 18

Final Link Budget (Baseline, Prediction and Allocation) Summary

		Operation	Units	Antenna Hardware Prediction (37 dB)	Requirements Allocation (35dB)	Baseline (31.5dB)	Margin = Prediction - Baseline	Margin = Prediction - Allocation
UAV Transmitter	Tx Power, Ptx	+	dBm	15.00	15.00	15.00	0.00	0.00
	Tx component line losses, L_T,Line	+	dB	-1.00	-1.00	-1.00	0.00	0.00
	Tx antenna gain, Gt	+	dBi	13.16	11.16	7.66	5.50	2.00
	Tx pointing loss, L_T,point	+	dB	-0.10	0.00	0.00	-0.10	-0.10
	Tx radome loss, L_T,radome	+	dB	-0.50	-1.00	-1.00	0.50	0.50
	EIRP		dBm	26.56	24.16	20.66	5.90	2.40
Propagation	Free space path loss	+	dB	-118.31	-118.31	-118.31	0.00	0.00
	Atospheric absorption, L_P,Atm	+	dB	-0.93	-0.93	-0.93	0.00	0.00
	Precipitation absorption, L_P, Precip	+	dB	-0.02	-0.02	-0.02	0.00	0.00
	Total Propagaton Loss		dB	-119.26	-119.26	-119.26	0.00	0.00
LCS-2 Control Station (Receiver)	Rx peak antenna gain, G_R	+	dBi	19.00	19.00	19.00	0.00	0.00
	Rx polarizaton loss, L_R, Polar	+	dB	-3.00	-3.00	-3.00	0.00	0.00
	Rx pointing Loss, L_R,Point	+	dB	0.00	0.00	0.00	0.00	0.00
	Rx radome loss, L_R, Radome	+	dB	0.00	0.00	0.00	0.00	0.00
	Rx component line losses, L_R, Line	+	dB	-1.00	-1.00	-1.00	0.00	0.00
	Spreading implemetation loss, L_R,Spread	+	dB	0.00	0.00	0.00	0.00	0.00
	Effective carrier power		dBm	-77.70	-80.10	-83.60	5.90	2.40
Noise	Thermal noise density, kT @ 290K	-	dBm/Hz	-174.00	-174.00	-174.00	0.00	0.00
	Rx noise bandwidth, BW	-	dBHz	50.79	50.79	50.79	0.00	0.00
	Rx noise figure, NF	-	db	4.00	4.00	4.00	0.00	0.00
	Effective noise power	-	dBm	-119.21	-119.21	-119.21	0.00	0.00
Summary	Available SNR		dB	41.51	39.11	35.61	5.90	2.40
	Required SNR		dB	4.11	4.11	4.11	0.00	0.00
	Net signal margin (link margin)		dB	**37.40**	**35.00**	**31.50**	**5.90**	**2.40**

7.3 Hardware Selection Summary

7.3.1 Airborne Transmitter Equipment

Component	**Specification**	**Rationale**
RF Transmitter	15.0 dBm output, 1060.5 MHz, GMSK modulation	Provides sufficient power to meet 24.16 dBm EIRP requirement
Power Amplifier	0.5 W, L-band, 80% efficiency	Maintains low power consumption while ensuring link closure
Dipole Antenna	13.155 dBi peak gain, omnidirectional horizontal pattern	Ensures continuous coverage during all bank angles; eliminates pointing requirements
Coaxial Cable	Low-loss LMR-400, 3 ft length, 0.5 dB/ft loss	Minimizes insertion losses to maintain EIRP

7.3.2 LCS-2 Ground Station Equipment

Component	**Specification**	**Rationale**
Parabolic Dish Antenna	19.0 dBi gain, 18° beamwidth, 1.5 m diameter	High gain ensures strong received signal; narrow beamwidth permits precise tracking
Gimbal System	2-axis, ±45° elevation, 360° azimuth, 5°/s slew rate	Enables rapid acquisition and continuous tracking of maneuvering UAV
LNA	Noise Figure ≤4.0 dB, L-band, 40 dB gain	Maintains low noise floor while boosting signal for demodulation
Demodulator	GMSK compatible, 138 kbps, $\leq 1\times10^{-6}$ BER	Meets data rate and error rate requirements for reliable link

7.4 Design Verification Summary

Requirement ID	**Description**	**Status**
CM-1	Maintain minimum Net Signal Margin of 35.0 dB	**Verified** (37.40 dB, achieved)
CM-2	BER of 10^{-6} at Eb/No threshold of 3.5 dB	**Verified**
CM-4	Transmitter Output Power ≥ 15.0 dBm	**Verified**
CM-5	UAV Antenna Gain ≥ 12.124 dBi (80% efficiency)	**Verified** (13.155 dBi selected)
CM-6	Minimum EIRP ≥ 24.16 dBm	**Verified** (26.55 dBm achieved)
CM-7	Control Station Antenna Gain ≥ 19.0 dBi	**Verified**
CM-8	Receiver Noise Figure ≤ 4.00 dB	**Verified**

6.0 Conclusion

The UAV communication system design has been successfully validated against all mission requirements for VTOL-based ISR operations from an LCS-2 ship in the South China Sea. The link budget analysis confirms that the system achieves a **net signal margin of 37.40 dB**, exceeding the 35.0 dB requirement by 2.40 dB. This margin provides adequate protection against environmental degradations such as tropical precipitation, atmospheric absorption, and potential RF interference in this contested electromagnetic environment.

The selected hardware configuration, a 15.0 dBm airborne transmitter with a 13.155 dBi omnidirectional antenna and a shipboard 19.0 dBi parabolic dish, satisfies all derived requirements while maintaining positive link closure at the 10 NM operational range. The dual-link architecture (LOS primary with BLOS backup) ensures continuous command and control across all mission phases, with seamless failover capability.

All key performance parameters, including BER ($\leq 1\times10^{-6}$), EIRP (26.55 dBm), and receiver noise figure (4.0 dB), meet or exceed their specified thresholds. The design demonstrates that a robust, reliable communication link can be established and maintained even under the challenging conditions of the South China Sea theater, enabling successful execution of the ISR mission.

References

Embry-Riddle Aeronautical University (n.d.). *Group Project #2: UAS Avionics, Autonomy, and Datalink Analysis* (UASE 501: Introduction to Unmanned Aircraft Design) [Course assignment]. Embry-Riddle Aeronautical University Canvas.

Federal Communications Commission. (2024). *Federal Code of Regulations: Online table of frequency allocations*. U.S. Department of Commerce. https://www.fcc.gov/sites/default/files/fcctable.pdf

International Telecommunication Union. (2014). *Guidelines for the preparation of a National Table of Frequency Allocations* (ITU-D Spectrum Management Publication). https://www.itu.int/en/ITU-D/Spectrum-Broadcasting/Documents/Publications/Guidelines-NTFA-E.pdf

ITU-R. (2005). *Specific attenuation model for rain for use in prediction methods* (Recommendation ITU-R P.838-3). International Telecommunication Union.

RTCA. (2016). *Command and control (C2) data link minimum operational performance standards (MOPS) (terrestrial)* (RTCA DO-362). RTCA Special Committee 228.

SBS News. (2024, June 9). *South China Sea dispute | SBS Explained* [Video]. YouTube. https://www.youtube.com/watch?v=sq_hj-bcYzw

Heywood, J. (2019, April 3). *China's economic cabbage strategy*. Asia Maritime Transparency Initiative. https://amti.csis.org/chinas-economic-cabbage-strategy/

Center for Maritime Strategy. (2024, June 20). *Deterring China's salami tactics*. https://centerformaritimestrategy.org/publications/south-china-sea-deterrence-project/

Council on Foreign Relations. (2026, May 14). *Territorial disputes in the South China Sea*. Global Conflict Tracker. https://www.cfr.org/global-conflict-tracker/conflict/territorial-disputes-south-china-sea

AIM Online. (n.d.). *AFDX®/ARINC664P7 Tutorial*. https://www.aim-online.com/products-overview/tutorials/afdx-arinc664p7-tutorial/

Young, L. A., Yetter, J. A., & Guynn, M. D. (2006, September 19). *System analysis applied to autonomy: Application to high-altitude long-endurance remotely operated aircraft*. National Aeronautics and Space Administration. https://ntrs.nasa.gov/citations/20090027657

The MathWorks. (n.d.). *Analyze coverage and view results*. MathWorks Help Center. Retrieved July 4, 2026, from https://www.mathworks.com/help/slcoverage/analyze-coverage-and-view-results.html

The MathWorks. (n.d.). *Model coverage*. MathWorks Help Center. Retrieved July 4, 2026, from https://www.mathworks.com/help/slcoverage/ug/model-coverage.html

The MathWorks. (n.d.). *WaypointTrajectory system object*. MathWorks Help Center. Retrieved July 4, 2026, from https://www.mathworks.com/help/nav/ref/waypointtrajectory-system-object.html